MA1522 Linear Algebra for Computing

Lesson 3: Matrices, Matrix Operations and Matrix Equations

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson3` (e.g. `A1234567Z_lesson3.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 3**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Basic rules:

- Please (mainly) ask the questions related to the course.

Section 2.1 Definition and Special Types of Matrices

An $m \times n$ **matrix** is a rectangular array $A = (a_{ij})_{m \times n}$ of m rows and n columns; a_{ij} is the entry in row i, column j — row index first, always. Its **size** is $m \times n$.

Two matrices are **equal** when they have the same size and the same entries in every position. An $n \times 1$ matrix is a **column vector** and a $1 \times n$ matrix a **row vector**; unqualified, "vector" means column vector.

If $m = n$ the matrix is **square**, of **order** n. Some square shapes:

- **diagonal**: $a_{ij} = 0$ whenever $i \neq j$; the **identity** I_n is the diagonal matrix with every $a_{ii} = 1$;
- **upper triangular**: $a_{ij} = 0$ whenever $i > j$; **lower triangular**: $a_{ij} = 0$ whenever $i < j$; **strictly** so if the diagonal is zero too;
- **symmetric**: $a_{ij} = a_{ji}$ for all i, j.

Section 2.2 Matrix Algebra: The Operations

- **Addition**: $A + B = (a_{ij} + b_{ij})_{m \times n}$ — same size required.
- **Scalar multiplication**: $cA = (c\, a_{ij})_{m \times n}$ — always defined.
- **Multiplication**: for A of size $m \times n$ and B of size $n \times p$,

$$AB = \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right)_{m \times p}.$$

- **Transpose**: $A^T = (a_{ji})_{n \times m}$ — reflect in the diagonal.
- **Power**: for A square only, $A^0 = I$ and $A^k = AA^{k-1}$ for $k \geq 1$; hence $A^k A^l = A^{k+l}$.



$\implies$ AB has size $m \times p$

Section 2.2 What Carries Over from Numbers, and What Does Not

Still true. For matrices of compatible sizes and $c \in \mathbb{R}$:

- $A + B = B + A$, $(A + B) + C = A + (B + C)$;
- $(AB)C = A(BC)$;
- $A(B + C) = AB + AC$, $(A + B)C = AC + BC$;
- $c(AB) = (cA)B = A(cB)$;
- $AI_n = A = I_m A$, $A + 0 = A$, $A0 = 0$;
- $(A + B)^T = A^T + B^T$, $(cA)^T = cA^T$;
- $(AB)^T = B^T A^T$.

No longer true. In general:

- $AB \neq BA$ — one may be defined and the other not;
- $(AB)^T \neq A^T B^T$ — the order reverses;
- $AB = 0$ does not force $A = 0$ or $B = 0$;
- hence $AB = AC$ does not force $B = C$.

Every cancellation habit you have from $\mathbb{R}$ needs re-examining.

Example 1

Let A and B be square matrices of order n. Which of the following are always true?

1. $(A + I_n)^2 = A^2 + 2A + I_n$.
2. $(A + B)^2 = A^2 + 2AB + B^2$.

(1) **True.** Expanding, $(A + I)^2 = A^2 + IA + AI + I^2 = A^2 + A + A + I$. This one survives because I commutes with *everything*.

(2) **False.** $(A + B)^2 = A^2 + AB + BA + B^2$, and $AB + BA \neq 2AB$ unless A and B commute. Concretely, with $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$,

$$(A + B)^2 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)^2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \qquad A^2 + 2AB + B^2 = 0 + 2\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) + 0 = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 0 \end{smallmatrix}\right).$$

The whole difference between (1) and (2) is whether the two matrices commute — which is exactly what Question 1 is about.

Question 1

Let $A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$.

1. Without computing anything, write down two matrices B with $AB = BA$.

2. Find all matrices B with $AB = BA$.

Question 1 — solution

(1) For AB and BA both to be defined, B must be 2×2. Two matrices commute with every square matrix of the right size, so no computation is needed:

$$B = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad B = 0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

(2) Write $B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. Then

$$AB = \begin{pmatrix} r & s \\ 2p+3r & 2q+3s \end{pmatrix}, \qquad BA = \begin{pmatrix} 2q & p+3q \\ 2s & r+3s \end{pmatrix}.$$

Comparing entries gives $r = 2q$ and $s = p + 3q$; the remaining two equations are then automatic. So p and q are free:

$$B = \begin{pmatrix} p & q \\ 2q & p+3q \end{pmatrix} = p \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + q \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} = pI_2 + qA, \qquad p, q \in \mathbb{R}.$$

Question 1 — remark

The answer $B = pI_2 + qA$ contains part (1): $p = 1, q = 0$ gives I_2, and $p = q = 0$ gives 0.

Now count parameters. A 2×2 matrix $\left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$ has **four** entries, each free to be anything, so *all* 2×2 matrices are described by four free variables — a copy of $\mathbb{R}^4$.

The ones commuting with A needed only **two**, p and q: the equation $AB = BA$ used up two of the four degrees of freedom. They form a copy of $\mathbb{R}^2$ sitting inside that $\mathbb{R}^4$.

So commuting with a fixed matrix is rare — a thin slice of the whole space, much as a plane is a thin slice of $\mathbb{R}^4$. Pick a B at random and expect $AB \neq BA$. (The exception is $A = cI$, which commutes with everything; then the slice is all of $\mathbb{R}^4$.) Chapter 3 makes "how many free variables" precise: it is the **dimension**.

Section 2.3 Linear Systems and Matrix Equations

A linear system in standard form is exactly the single **matrix equation**

$$A\mathbf{x} = \mathbf{b}, \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix},$$

with **coefficient matrix** $A = (a_{ij})_{m \times n}$, **variable vector** $\mathbf{x}$ and **constant vector** $\mathbf{b}$.

Writing $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix}$ by columns — $\mathbf{a}_i$ is the **coefficient vector** of x_i — this is the **vector equation**

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}.$$

Question 2

Consider the linear system

$$\begin{cases} x_1 + 2x_2 + x_3 + 3x_4 = 8 \\ x_1 - 2x_2 - x_3 + x_4 = -2 \\ 2x_1 - x_2 - x_3 + 7x_4 = 1 \end{cases}$$

1. Write it as a matrix equation $A\mathbf{x} = \mathbf{b}$ and as a vector equation.
2. Solve it, and write the general solution in the form $\mathbf{u} + t\mathbf{v}$.
3. Write $A = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{pmatrix}$ by columns. Hence find real numbers c_1, c_2, c_3 with

$$\mathbf{a}_4 = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3.$$

Question 2 — solution: parts (1) and (2)

$$\underbrace{\begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & -2 & -1 & 1 \\ 2 & -1 & -1 & 7 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 1 \end{pmatrix}, \quad x_1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + x_4 \begin{pmatrix} 3 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 1 \end{pmatrix}.$$

Gauss–Jordan on the augmented matrix gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 8 \\ 1 & -2 & -1 & 1 & -2 \\ 2 & -1 & -1 & 7 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 4 & 0 \\ 0 & 0 & 1 & -7 & 5 \end{array}\right)$$

Columns $1, 2, 3$ are pivot and column 4 is not, so put $x_4 = t$:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 \\ 0 \\ 5 \\ 0 \end{pmatrix}}_{\mathbf{u}} + t \underbrace{\begin{pmatrix} -2 \\ -4 \\ 7 \\ 1 \end{pmatrix}}_{\mathbf{v}}, \qquad t \in \mathbb{R}.$$

Question 2 — solution: part (3)

Substituting $\mathbf{u} + t\mathbf{v}$ into $A\mathbf{x} = \mathbf{b}$ gives $A\mathbf{u} + t\,A\mathbf{v} = \mathbf{b}$ for *every* t, so

$$A\mathbf{u} = \mathbf{b} \qquad \text{and} \qquad A\mathbf{v} = \mathbf{0}.$$

Now read $A\mathbf{v} = \mathbf{0}$ as a vector equation. The entries of $\mathbf{v}$ are the coefficients:

$$-2\mathbf{a}_1 - 4\mathbf{a}_2 + 7\mathbf{a}_3 + 1 \cdot \mathbf{a}_4 = \mathbf{0}, \qquad \text{so} \qquad \mathbf{a}_4 = 2\mathbf{a}_1 + 4\mathbf{a}_2 - 7\mathbf{a}_3,$$

that is, $c_1 = 2$, $c_2 = 4$ and $c_3 = -7$.

Check: $2\begin{pmatrix}1\\1\\2\end{pmatrix} + 4\begin{pmatrix}2\\-2\\-1\end{pmatrix} - 7\begin{pmatrix}1\\-1\\-1\end{pmatrix} = \begin{pmatrix}2+8-7\\2-8+7\\4-4+7\end{pmatrix} = \begin{pmatrix}3\\1\\7\end{pmatrix} = \mathbf{a}_4$. ✓

Note that c_1, c_2, c_3 can also be read straight off column 4 of the RREF, namely $\begin{pmatrix}2\\4\\-7\end{pmatrix}$: **a non-pivot column of the RREF records exactly how the corresponding column of A is built from the pivot columns before it.** □

Section 2.3 Block Multiplication

The same idea, one column at a time. If $B = \begin{pmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{pmatrix}$ has columns $\mathbf{b}_j$ and A has rows $\mathbf{r}_1, \ldots, \mathbf{r}_m$, then

$$AB = A \begin{pmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{pmatrix} = \begin{pmatrix} A\mathbf{b}_1 & \cdots & A\mathbf{b}_p \end{pmatrix}, \qquad AB = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \end{pmatrix} B = \begin{pmatrix} \mathbf{r}_1 B \\ \vdots \\ \mathbf{r}_m B \end{pmatrix}.$$

j-th column of $AB = A \times (\, j$-th column of $B)$; $\quad i$-th row of $AB = (\, i$-th row of $A) \times B$.

Question 3

Let $A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$.

Find a 4×3 matrix X with $AX = I_3$. Is it unique?

Question 3 — solution

Writing $X = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 \end{pmatrix}$, block multiplication turns $AX = I_3$ into three systems $A\mathbf{x}_j = \mathbf{e}_j$ with the *same* coefficient matrix, so one Gauss–Jordan run on $(A \mid I_3)$ does all three:

$$\left(\begin{array}{cccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 2 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

Column 4 is non-pivot, so each $\mathbf{x}_j$ carries one parameter. Taking all three to be 0:

$$X = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad AX = I_3 \ \checkmark$$

It is not unique — there are infinitely many such X.

Challenge

Question 3 produced infinitely many X with $AX = I_3$. Explain why, for the same A, no 4×3 matrix Y satisfies $YA = I_4$.

There are various ways to argue this. The best one goes through the concept of **rank** in Section 4.2; here is an alternative using only what we have now.

Suppose there exists such a Y. Take any X with $AX = I_3$. Then

$$Y = YI_3 = Y(AX) = (YA)X = I_4X = X.$$

So Y equals *every* such X. But Question 3 produced infinitely many different choices of X, so this equality cannot hold.

One-sided inverses are the subject of Section 2.4.

Next lesson

- **Sections 2.4–2.6 Inverses, Elementary Matrices and Their Uses**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.